

Hop Hop Hop! Design Report

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Authored by: Ciara Looney



Table of Contents

Top-Level Design Description.....	3
Design Pillars	3
Audience & Marketing.....	3
Core Gameplay	3
Controls.....	4
Gameplay Balance & Pacing.....	4
Tone & Aesthetics.....	5
Narrative	6
Business Model.....	6

Top-Level Design Description

Hop Hop Hop! is a survival game where the player must hop around the lilypads and eat flies. The goal is to last as long as possible. At the start of the game players can stand on a lilypad for 3 seconds before it breaks but later on in the game they break sooner. If a player is standing on a lilypad as it breaks, they lose a life. Players can also eat flies to get extra lives. The game ends when the player runs out of lives.

Design Pillars

The below design pillars identify the types of fun which are key to the user experience

Thrill of Danger	Application of an Ability	Intellectual Problem Solving
Once you land on a pad it starts to sink. If you stay too long it may cost you a life but if you move too quickly you may go in the wrong direction. Since the lilypads break more quickly as the game goes on the risk never goes down.	You are hopping in a grid-based map. The tongue attack is aim-based as its tied to the direction the frog is facing. So, in order to eat a fly, it comes down to precision as you must aim correctly while the lilypad doesn't break underneath it.	Progress is tracked through a timer. There is also a bronze, silver and gold tier. Every run clearly tells you what you achieved and what you could do if you did a bit better next time.

Audience & Marketing

This is designed to be an offline game played via desktop. It is a single player survival game. It will be aimed at a casual arcade audience who may like short, replayable sessions with a clear high score to chase.

Core Gameplay

The player starts with 3 lives and several lilypads surrounding them. They can press any direction to hop to another lilypad. Lilypads sink shortly after the player lands on them

so returning to a previous pad is not possible. If the player stays on a lilypad too long after it starts sinking, they lose a life. Flies periodically appear on pads and the player can face them and eat them, earning them an extra life.

The player tries to last as long as possible which is tracked by a timer. As the timer progresses, the time before a lilypad sinks decreases. This makes hops more frequent and increases the chance of the player getting stuck as they couldn't plan a path. The goal is to last as long as possible. The timer is divided into bronze, silver and gold tiers which gives the player a target to beat on each run.



Controls

WASD / Arrow Keys – Aim/Hop one lilypad in that direction

Space – Shoot your tongue

Gameplay Balance & Pacing

The game's difficulty curve is built directly into the core mechanic. Every pad you land on immediately starts breaking. This gives the player 3 seconds to move on before it sinks. This threshold shrinks as the game goes on. It goes down to as low as 1.2 seconds. So players at the start of the game have time to learn the mechanic while the

late game tests how well players can react with much less time. The lowest time at 1.2 is difficult but not unplayable.

An early version of the game tracked score as pads hopped, but nothing pushed the player forward and a run could continue indefinitely with no reason to continue. As the game changed the pads also became a danger. This is communicated clearly to the player as pads darken and shrivel as soon as the player lands on one. So a player shouldn't be surprised to lose a life this way. But if they do a player will respawn where they fell rather instead of being sent back to the start. This keeps the game moving forward.

Pads also sink behind the player once they hop off, which stops the player from bouncing safely between two pads indefinitely. Pads that were destroyed will respawn after a while. This will give the player other paths to explore and not leave them stranded. Flies add risk and reward as each fly restores a life but they can be dangerous to catch. If a player wants to catch a fly they may need to detour. The player must carefully line up the attack as they can't hop or change directions mid-attack. So when the player begins their attack the pad they're standing will begin to break while they commit to the catch so they must ensure they have begun their attack shortly after landing on the lilypad or else they may lose a life.

Progress is measured by how long the player can survive. The player's time is shown on the game over screen. If they last a long time they may also be split into a bronze, silver or a gold tier. The use of tier is to encourage the player to try again to reach the next tier. For example, if the player landed in the silver tier they may be tempted to try again.

Tone & Aesthetics

The visual style is a pastel low-pixel pixel art. I styled it around the frog sprite I had purchased. So, I drew the pads and water at a similarly low pixel density so the sprite didn't look out of place.

The color palette is pastel throughout. It has pastel blue water and pastel green lily pads keeping the pond calm and inviting despite the constant pressure of the game. I had several coloured frogs to choose from and I chose the red frog which doesn't fit the pastel palette on purpose so the player can always spot it instantly against the pond regardless of how much is on the screen.

Narrative

This is an offline singleplayer game without any explicit narrative. I do not intend to have dialog or characters speaking.

Business Model

Smooth Snake is a free-to-download Nintendo DS game with no monetisation model. It will be made freely available to all users with no ads and no premium tiers.

The game is being developed as a portfolio project to demonstrate skills in game design and development. Its primary value is in the player experience and the learning gained through building it, rather than revenue generation.